

Review: AstroAlertBench

Chen, C., Xiao, J. S., Liu, S. D., Perez Paolino, F., Handley, L., Jegou du Laz, T., Nilsson, R., Zou, A., Graham, M. & Mahabal, A. (2026), “AstroAlertBench: Evaluating the Accuracy, Reasoning, and Honesty of Multimodal LLMs in Astronomical Classification”, arXiv:2605.05573v1 [astro-ph.IM].

Summary

AstroAlertBench is a 1,500-alert multimodal benchmark (300 each of bogus, asteroid, supernova, AGN, variable star) built from ZTF first-detection alerts served through ALERCE. Each item is a labelled science/reference/difference montage of three 63x63 cutouts plus 19 AVRO metadata fields; the returned JSON has Part A (six decoded fields), Part B (three rationale fields, self-scored 0-5) and Part C (a cascade: artifact, origin, subclass). Thirteen configurations from five families run one sample per alert at temperature 0.2 where the API accepts it - GPT-5.4 and Opus 4.7, the two leading families, omit it (App. A.1, uncaveated in Table 2). Claude Opus 4.7 think leads end-to-end at 60.60 +/- 1.26%, 9.53 pp above GPT-5.4 high-think. The distinctive material is the three-lens “honesty” analysis and a small human cohort (5 experts, 15 alerts, 75 trials, abstention permitted) - not the leaderboard.

Main contributions

1. **A staged protocol that separates instrument plumbing from science.** Every run is reported at 100.00 +/- 0.00% on all six Part A fields (Sect. 4.2, App. C.1), which the authors take as proof that downstream errors are not serialization failures - see below. The cascade then localises failure to a named gate: Stage-2 spans 54.21-81.60%, Stage-3 conditional accuracy 32.12-60.78% (Table 1).
2. **A reasoning-mode ablation on identical alert sets, analysed with the wrong test.** Five same-architecture pairs at $n = 1,500$ give Opus 4.7 +11.73 +/- 1.80 pp ($z = 6.52$), Qwen3.5-397B +9.29 (5.25), GPT-5.4 +7.40 (4.07), a tie at 35B (+1.23) and an inversion at 4B (-13.90, truncation) (Table 13). Both arms see the same alerts, yet App. D propagates the independent two-sample $SE_{\text{delta}} = \sqrt{SE_{\text{think}}^2 + SE_{\text{nothink}}^2}$ instead of McNemar on the discordant cells, which the same authors use correctly in Table 17; the quoted z are conservative, not correct.
3. **A universal, model-independent AGN failure.** AGN accuracy stays below 8% and AGN F1 below 0.13 in all 13 runs (Tables 9, 11); in the four leading runs 272-278 of the 300 true AGN are predicted variable_star (Table 12). No per-class lead reaches $z = 2$ (Table 10) - unusually honest leaderboard reporting.
4. **An inverse modesty-accuracy relation, then its instance-level refutation.** An OLS fit across the 13 runs gives -31.2 +/- 10.0 pp of accuracy per rubric point (Fig. 5): Opus at MSRS 3.99 and a 67% self pass rate, Gemini 2.5 Pro at 4.886 and 100%. But per-alert the confidence-correctness Pearson r never exceeds 0.2517 (Opus nothink), and adaptive thinking on Opus raises accuracy while collapsing that signal to $r = 0.047$ and a +0.24 pp bin gap (Tables 14, 15).
5. **A behavioural second-rollout test.** Re-prompting five of the six closed-source runs (Gemini 2.5 Pro high-think has no low-confidence pool, $n \leq 2$, Table 15) on 35 of each model’s own low-confidence alerts, first answer attached, gives Opus think +28.57 pp (correction 58.82%, damage 0%, McNemar $p = 0.00195$), Opus nothink +42.86 pp (68.18%, 0%, $p = 6.1e-5$), GPT-5.4 high-think +8.57 pp while breaking 18.75% of correct rows ($p = 0.51$), and Gemini 2.5 Flash changing nothing while raising its self-rating +0.43 (Table 16).

Critical assessment

Stage-1 is not “essentially solved across all model scales (77.15-87.07%)” (Fig. 8); it is at the majority-class floor. With 300 bogus among 1,500 alerts, always answering real_object scores exactly 80.00%; four runs fall below that (77.15-79.30%), and only Opus 4.7 think clears it by 5 pp (86.60%). The paper names the mechanism itself: “a strong majority-class bias in Stage-1” (Sect. 4.4). Stage-3 conditional accuracy likewise does not “explain most of the headline 5-class ranking” (Fig. 8): its Spearman rho against end-to-end rank is 0.588, below Stage-2’s 0.659, and Opus 4.7 nothink is second on it, fourth end-to-end.

The central problem is the label. Gold classes are ALerCE stamp-classifier outputs drawn from a narrow slice of that classifier’s confidence surface (AGN 0.860-0.899, asteroid exactly 1.000; Table 6), so the benchmark measures agreement with a CNN. That reframes the Conclusion’s claim that frontier models “can exceed human classification accuracy” (Sect. 6) into the weaker statement that they imitate that classifier better than five astronomers do. Credit where due: App. G.1 concedes the expert-vs-manifest gap “is itself a signal”, and G.4 reads many human misses as calibrated refusal. That caveat belongs beside Table 1’s human row and Fig. 21’s caption, not alone in Appendix G.

The human comparison cannot bear its weight (Table 19):

Same 15 alerts	Accuracy (%)	n
Claude Opus 4.7 think	53.33 +/- 12.88	15
Best individual expert	46.67 +/- 12.88	15
Effective human (abstain wrong)	30.67 +/- 5.32	75
Strict 3/5 ensemble	26.67 +/- 11.42	15

The model-beats-expert result is 8 correct versus 7, one alert worth 6.67 pp. Per-expert accuracy ranges 13.33-46.67% with no calibration round, three of the five aliases match author names, and humans may answer “Don’t Know” (10.67% refusal) while models are forced to commit. The slice is too small to rank anything.

Part A is a null test, not a perception test: four numbers copied from the prompt survive a round trip, and two coded fields (fid, isdiffpos) are decoded from a key the prompt supplies. The grounding failure it should catch sits in the authors’ own material: on ZTF26aargnnp Gemini 2.5 Flash calls the source “absent in the reference” in one field of a JSON and “clearly present in the reference” in another, self-scoring them 5 and 3 (App. H.3.3) - which no coherence rubric over a model’s prose can catch.

The second-rollout design lacks its control arm: the low-confidence stratum is selected on a within-model extreme, so regression to the mean alone predicts a positive delta, and the +28.57/+42.86 pp gains cannot be credited to self-critique without a prior-withheld re-run. Parse failures also give one Table 1 row two denominators: Qwen3.5-4B think’s Stage-1 bar (87.07 +/- 1.88) inverts to $n \sim 318$ - Table 14’s $n_{\text{linked}} = 317$ - while its 5-class bar (8.41 +/- 0.72) inverts to $n \sim 1,486$; Qwen3.5-35B think splits 967 against $\sim 1,500$. Unparsed rows are dropped from the stage columns but scored wrong end-to-end, so the caption’s “Results are reported over 1,500 alerts” is false for two rows.

Relation to the agentic lensing programme

First, external corroboration of our label-wall thesis. Their five experts agree with the manifest gold on 30.67% of 75 trials and the best model reaches 53.33% on the same 15 alerts, while the headline 60.60% belongs to a different, 1,500-alert population. We have the same asymmetry from the other side: identical engineered features score AUC ~ 0.80 on spectroscopically confirmed labels

and ~ 0.51 on the DESI hard consensus pool, inter-team QWK is 0.29 [0.12-0.44] ($n = 103$) and 0.17 ($n = 24$), and the human grade’s own AUC against followed-up truth is 0.577 [0.40, 0.76]. Cite it in LensJudge that way: a benchmark’s headline is bounded by its labels, not by LLM astrophysics competence.

Second, it caps what abstention buys, lower than we assumed: selective accuracy exceeds effective by only 3.7 pp (34.33 $\pm$ 5.80 on 67 committed trials against 30.67 $\pm$ 5.32 on 75), a gap inside its own error bars that App. G.4 calls “small relative to the cross-model spread”, and measured on humans only - their models are forced-choice. In our JWST three-persona run uncertain was effectively dead (5 of 1,050 judgements); the evidence-first critics address that by design with an explicit no_opinion, but we have not published the rate at which they use it. We should generalise their three-definition reporting into the coverage-accuracy curve they stop short of.

Third, and this is what we should actually adopt: run their second-rollout ablation on our advocate. Our holdout makes the Opus-5 advocate the best ranker we have measured (recall@5%FPR 33.3% vs incumbent 9.5%, AUC 0.764, McNemar p 0.046), but the full critic stack collapses that ranking (AUC 0.468, dAUC -0.301) while remaining the right FPR layer (D-rule-only veto R2, FPR@A-union-B 2.0%). A same-model retry on the advocate’s own low-p_ev rows costs one extra call, against the four-judge panel mode’s $\sim$ \$0.27/candidate (multiagent $\sim$ \$0.33), and stacks no correlated refuters, though it inherits the advocate’s priors, so run a prior-withheld control arm beside it, both scored by dAUC and recall@5%FPR. Their per-gate conditional accuracy is worth stealing too: it would have exposed our geometry hard gate (2.3% pass) as one arm of the cascade that, with correlated refuters (kappa 0.46-0.63) and “prefer fail” mandates, rejected 23 of 24 known lenses.

Where it will not transfer: the operating point. Lenses are ~ 1 in 10^4 massive ellipticals, so balanced five-class accuracy above a 20% chance floor says nothing about purity at that prior; our currency is recall@5%FPR and FPR control. The honesty machinery transfers less well still: Part B scores prose against a coherence rubric, but our diagnosed failure was priors hard-coded wrong for NIRCcam - theta_E “0.3-3 arcsec, larger implies group/cluster”, and “the arc is bluer than the deflector”, which fires on 62% of real lenses. That reasoning is coherent and evidence-citing; a self-score rubric would rate it 5/5. And this is prompted classification, not agentic work - no tool use, no code execution - against a JWST run that designed its own pipeline over 6,848 NIRCcam observations and inspected 5,308 cutouts from one prompt.

Bottom line

This is a careful, well-instrumented benchmark whose honesty analysis is more novel than its leaderboard. Its structural weaknesses are gold labels that are a CNN’s opinion from a narrow probability band, and a human comparison resting on 15 alerts, five graders and an asymmetric abstention rule. The abstract is appropriately modest; the overreach is downstream, in the Conclusion’s “can exceed human classification accuracy”, Table 1’s human row and Fig. 21. Recommend acceptance conditional on demoting the human-parity claim, printing Stage-1’s 80.00% majority-class baseline, replacing the two-sample SE with McNemar, and adding a prior-withheld control to the retry ablation.